

LECTURE 6

Elementary Fluid Dynamics: The Bernoulli Equation

Reference: Munson, Rothmayer, Okiishi & Huebsch, Fundamentals of Fluid Mechanics, Chapter 3
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Inviscid flow and Newton's second law

We consider the motion of inviscid fluids: the viscosity is taken to be zero.

No real fluid is inviscid, since every fluid supports a shearing stress when it is strained.

But in many flows the viscous forces are orders of magnitude smaller than the pressure pressure and gravity forces and ignoring them is a good first approximation.

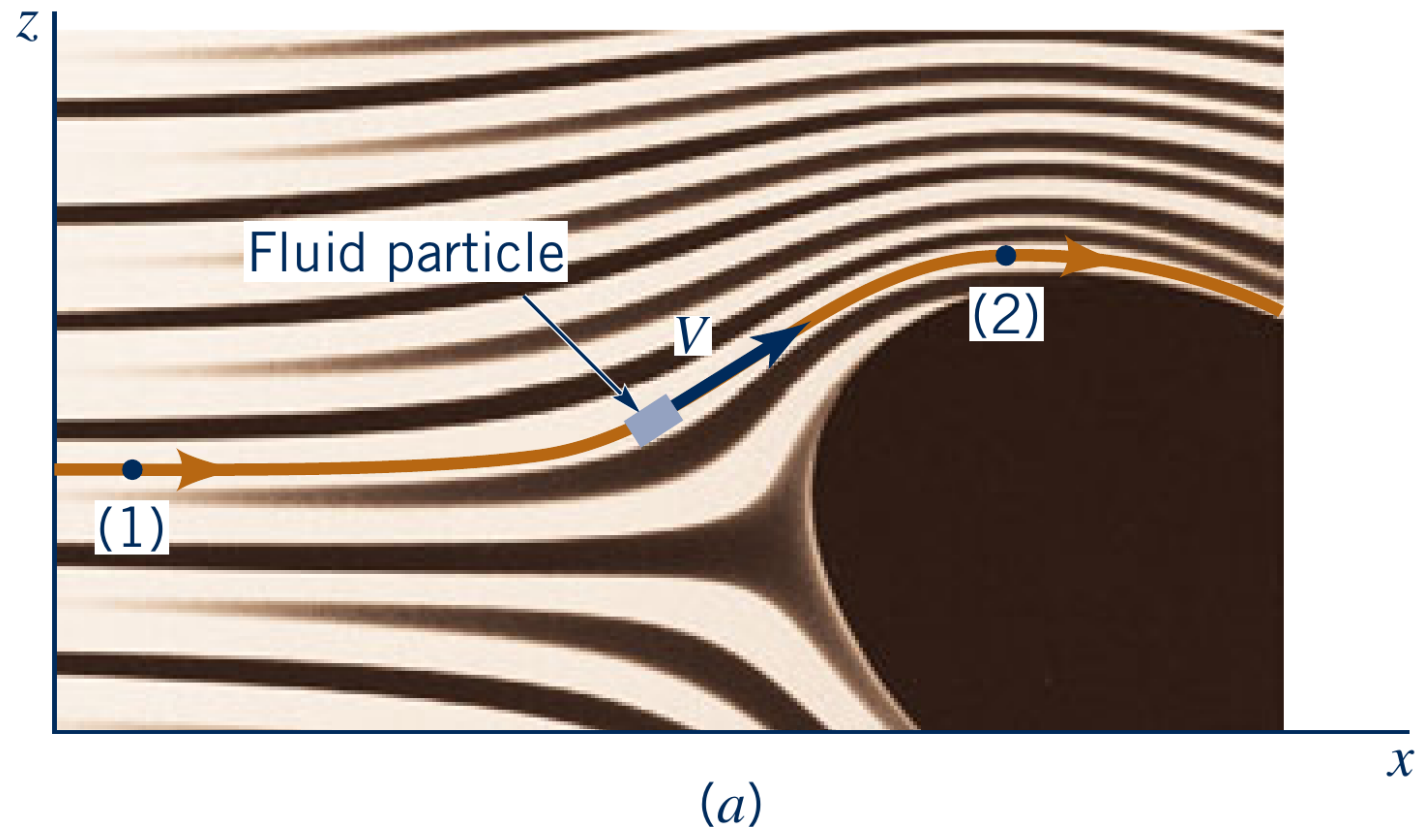
The whole unit is Newton's second law applied to a fluid particle, with only 2 forces kept:
kept:

$$\begin{aligned} & (\text{net pressure force on particle}) + (\text{net gravity force on particle}) \\ & = (\text{particle mass}) \times (\text{particle acceleration}) \end{aligned}$$

The Bernoulli equation may be the most used and abused equation in fluid mechanics.

Newton's second law

As a fluid particle moves from one place to another it usually accelerates. The net force on it must equal its mass times its acceleration:



$$\mathbf{F} = m \mathbf{a}$$

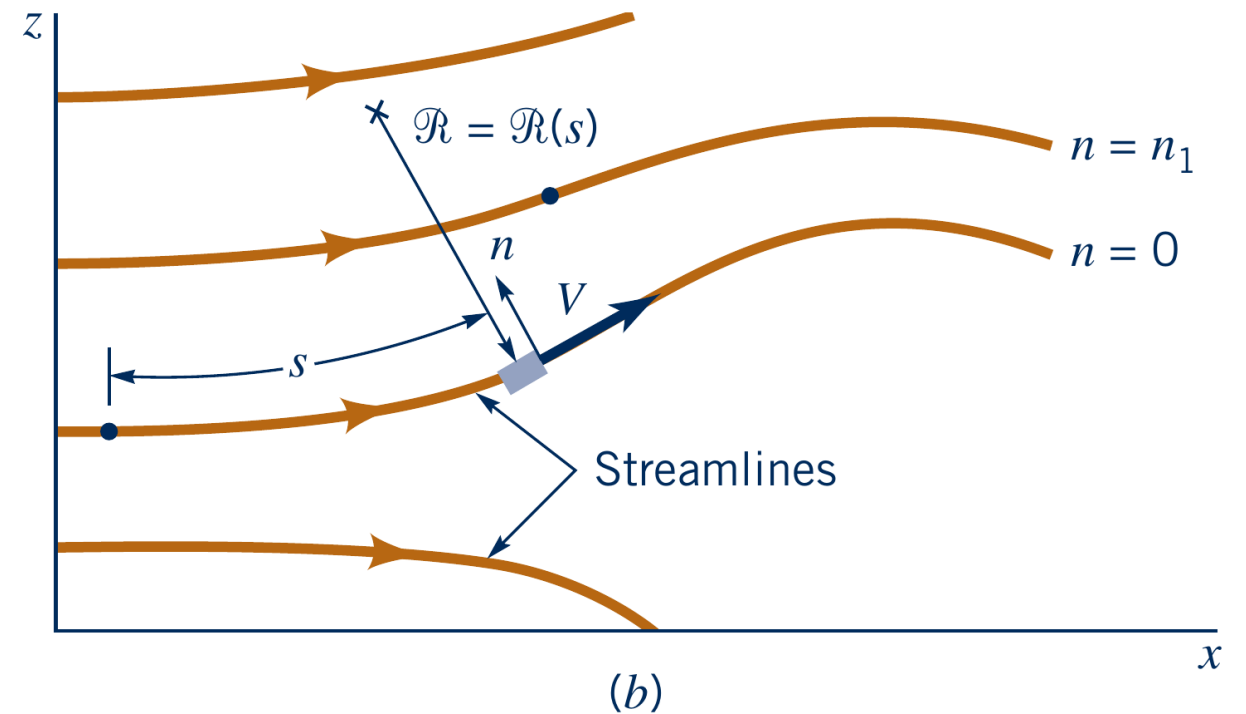
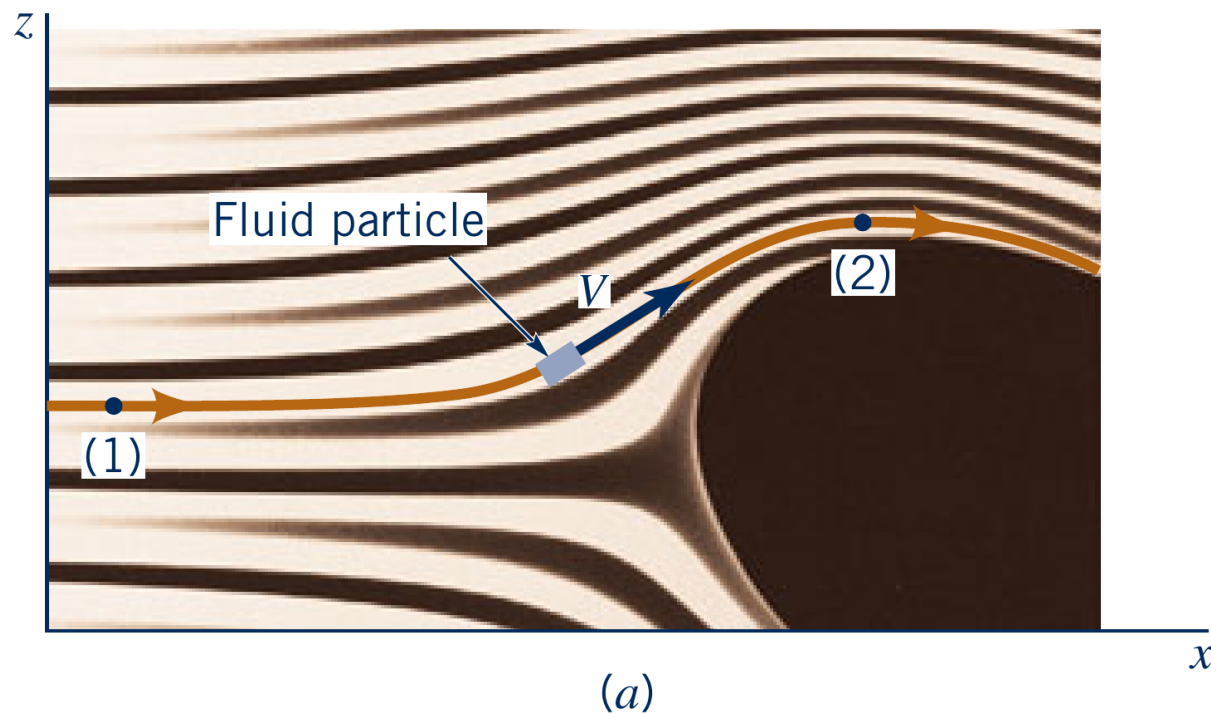
We assume the motion is governed by **pressure** and **gravity** alone, and we work in the x - z plane.

The particle velocity v is the time rate of change of its position, and the **lines tangent to the velocity vectors throughout the flow field are the streamlines**.

Inviscid fluid flow is governed by pressure and gravity forces.

Streamline coordinates

Instead of x and z , we describe the particle by its distance s along the streamline and by the local radius of curvature $R = R(s)$. The coordinate n is normal to the streamline.



(a) Flow in the x - z plane. (b) Flow in terms of streamline and normal coordinates.

$$v = \frac{ds}{dt}, \quad \mathbf{a} = \frac{d\mathbf{v}}{dt}$$

Fluid particles accelerate normal to and along streamlines.

The two components of the acceleration

The speed generally varies along the streamline, $V = V(s)$. By the chain rule, the **streamwise acceleration** is:

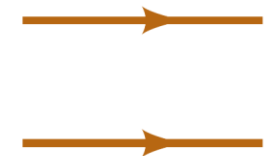
$$a_s = \frac{dv}{dt} = \frac{\partial v}{\partial s} \frac{ds}{dt} = \frac{\partial v}{\partial s} v$$

Since $\partial v / \partial s$ may be positive, negative or zero, the particle may accelerate, decelerate or hold its speed.

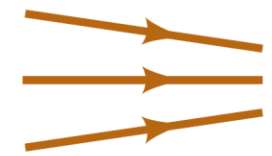
The normal component is the **centrifugal acceleration**, set by the speed and by the radius of curvature of the path:

$$a_s = v \frac{\partial v}{\partial s}, \quad a_n = \frac{v^2}{\mathcal{R}}$$

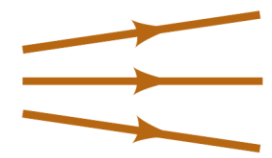
Converging streamlines produce a positive streamwise acceleration. To produce it there must be a net force on the particle.



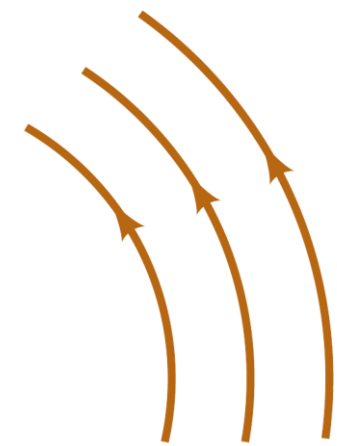
$$a_s = a_n = 0$$



$$a_s > 0$$



$$a_s < 0$$



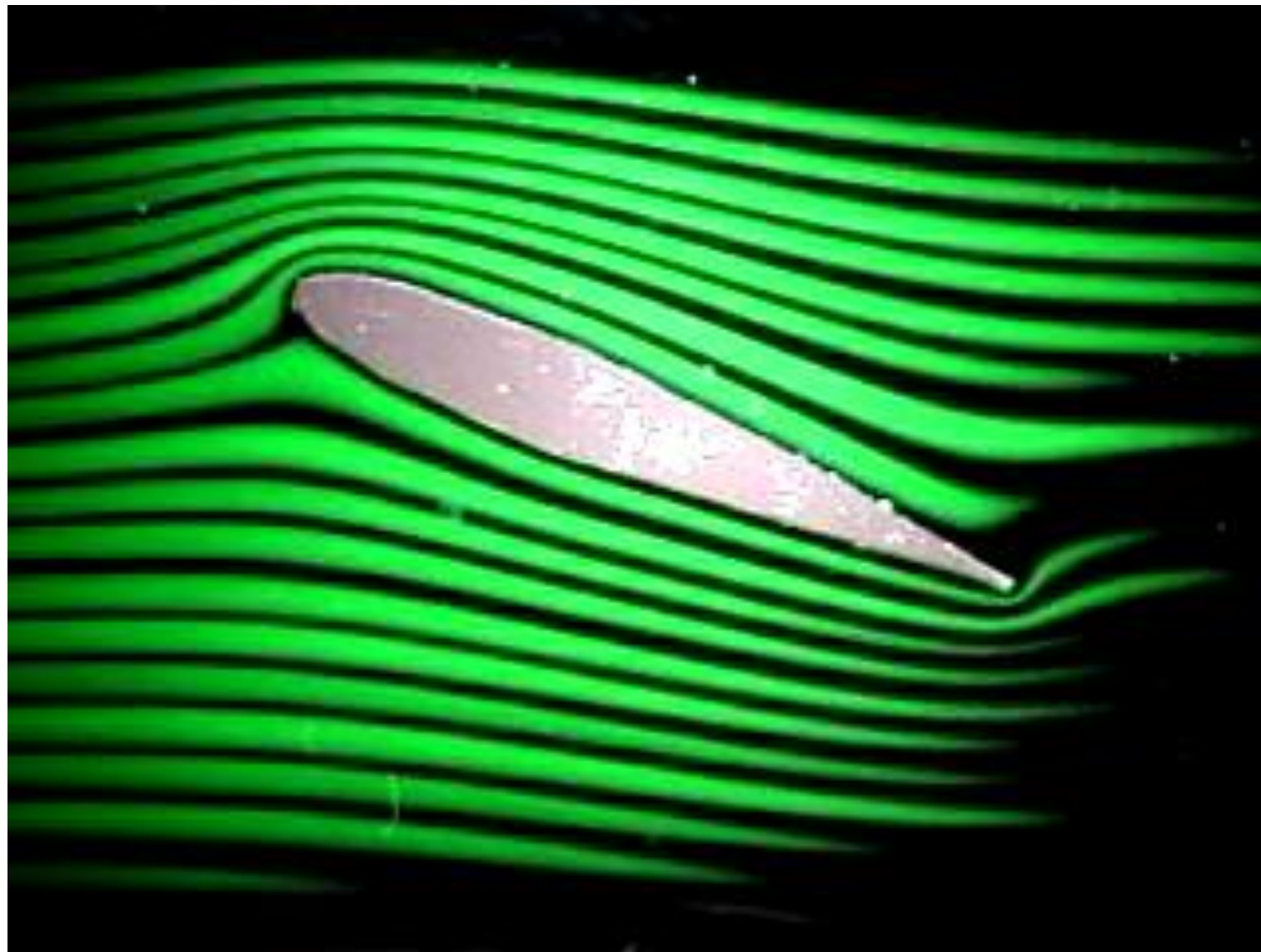
$$a_n > 0$$



$$a_s > 0, a_n > 0$$

Streamlines past an airfoil

Streamlines are lines everywhere tangent to the fluid velocity. For steady flow they coincide with the pathlines the fluid particles follow, so a streamline is the same as the path taken by dye injected into the flow.

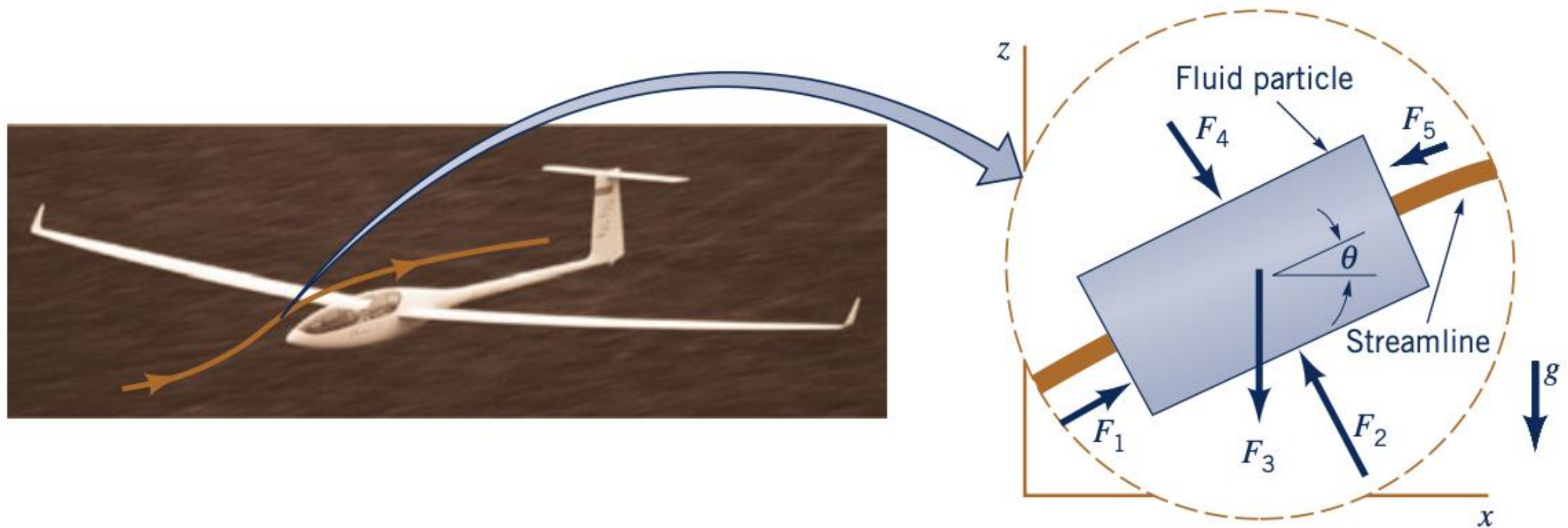


Video © S. T. Thoroddsen & Stanford University

The free-body diagram of a fluid particle

To find the forces that produce a given flow we isolate a small fluid particle and replace its surroundings by the forces they exert.

Only gravity and pressure are kept; viscous and surface-tension forces are assumed negligible. Gravity is constant, acts in the negative z direction, and makes an angle θ with the normal to the streamline.



Isolation of a small fluid particle in a flow field. (Photo courtesy of Diana Sailplanes.)

$F = ma$ along a streamline

Take a particle of size δs by δn in the plane of the figure and δy normal to it. For steady flow, the streamwise component of Newton's second law is

$$\sum \delta F_s = \delta m a_s = \delta m v \frac{\partial v}{\partial s} = \rho \delta \mathcal{V} v \frac{\partial v}{\partial s}$$

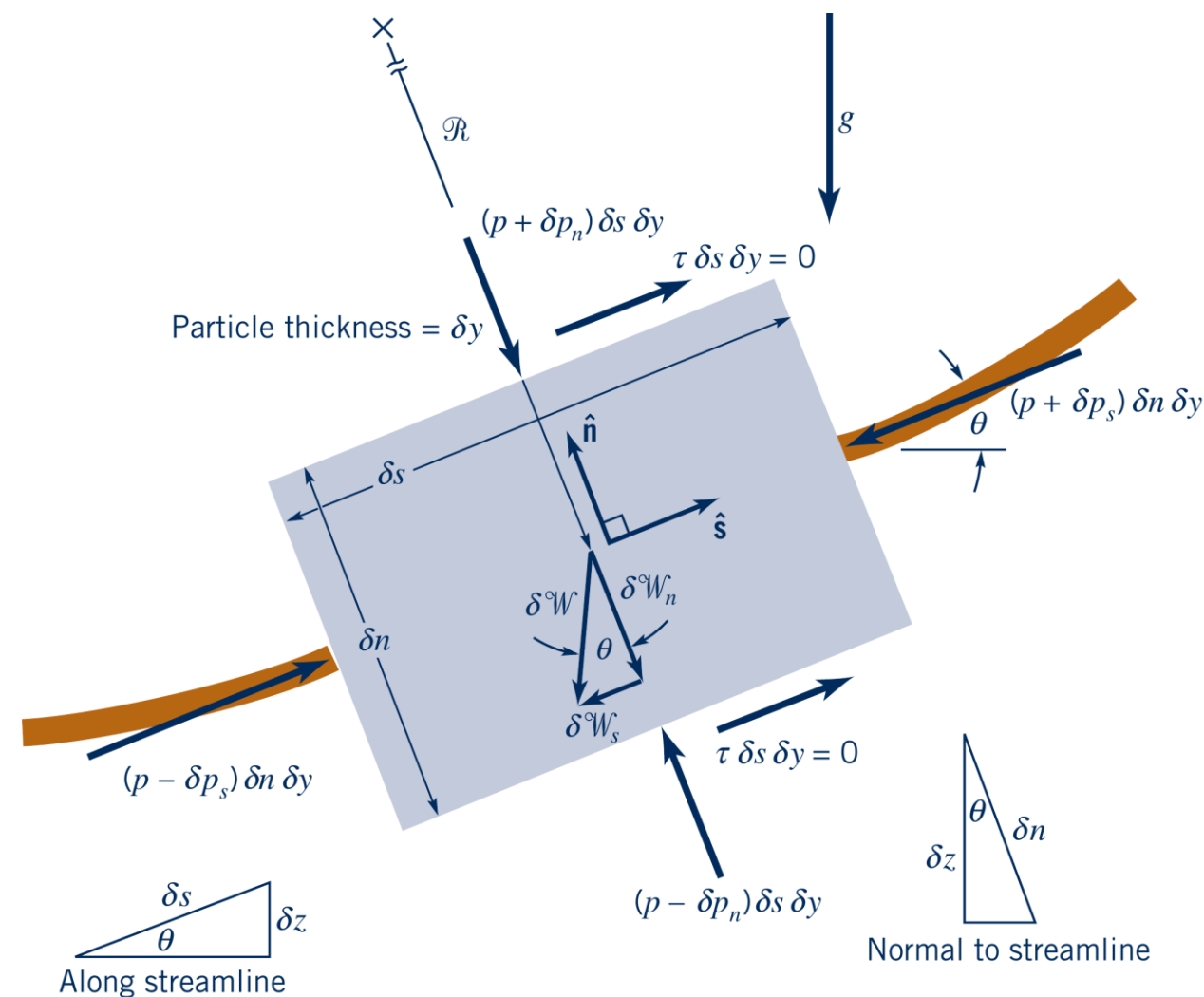
with mass $\delta m = \rho \delta \mathcal{V}$ and volume $\delta \mathcal{V} = \delta s \delta n \delta y$.

This holds for compressible and incompressible fluids alike:
the density need not be constant.

The component of the weight along the streamline is

$$\delta \mathcal{W}_s = -\delta \mathcal{W} \sin \theta = -\gamma \delta \mathcal{V} \sin \theta$$

so a horizontal streamline, $\theta = 0$, has no weight contribution in that direction.



Free-body diagram of a fluid particle for which the important forces are those due to pressure and gravity.

The net pressure force

In a flowing fluid the pressure is generally not constant, $p = p(s, n)$. If the pressure at the centre of the particle is p , a one-term Taylor expansion gives its average value on the two end faces as $p \pm \delta p_s$, with

$$\delta p_s \simeq \frac{\partial p}{\partial s} \frac{\delta s}{2}$$

so that the net pressure force along the streamline is

$$\begin{aligned} \delta F_{ps} &= (p - \delta p_s) \delta n \delta y - (p + \delta p_s) \delta n \delta y \\ &= -\frac{\partial p}{\partial s} \delta s \delta n \delta y = -\frac{\partial p}{\partial s} \delta \mathcal{V} \end{aligned}$$

The level of the pressure is irrelevant. What produces a net force is the fact that the pressure is not constant: **the nonzero pressure gradient**

$$\nabla p = \frac{\partial p}{\partial s} \hat{s} + \frac{\partial p}{\partial n} \hat{n}$$

The net pressure force on a particle is determined by the pressure gradient.

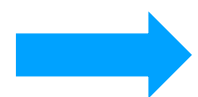
The equation of motion along the streamline

Adding the weight and the pressure contributions,

$$\sum \delta F_s = \delta \mathcal{W}_s + \delta F_{ps} = \left(-\gamma \sin \theta - \frac{\partial p}{\partial s} \right) \delta \mathcal{V}$$

and combining this with Newton's second law gives

$$-\gamma \sin \theta - \frac{\partial p}{\partial s} = \rho v \frac{\partial v}{\partial s} = \rho a_s$$



A change in speed comes from pressure gradient and weight along the streamline

The common factor $\delta \mathcal{V}$ has divided out: it is the density, not the mass of the particle, that matters.

In a static fluid the two terms on the left balance and nothing accelerates. In a flowing fluid they need not balance, and the imbalance is the motion.

Integrating along the streamline

Along the streamline:

$$\sin \theta = \frac{dz}{ds}, \quad v \frac{dv}{ds} = \frac{1}{2} \frac{d(v^2)}{ds}, \quad \frac{\partial p}{\partial s} = \frac{dp}{ds}$$

With these, our equation becomes:

$$-\gamma \sin \theta - \frac{\partial p}{\partial s} = \rho v \frac{\partial v}{\partial s} = \rho a_s$$

$$\Rightarrow dp + \frac{1}{2} \rho d(v^2) + \gamma dz = 0 \quad (\text{along a streamline})$$

and for constant g this integrates to

$$\int \frac{dp}{\rho} + \frac{1}{2} v^2 + gz = C \quad (\text{along a streamline})$$

For steady, inviscid flow the sum of certain pressure, velocity and elevation effects is constant along a streamline.

The Bernoulli equation

Assume in addition that the density is constant — a very good assumption for liquids, and for gases when the speed is not too high. Then our equation becomes:

$$p + \frac{1}{2}\rho v^2 + \gamma z = \text{constant along a streamline}$$

This is the **Bernoulli equation**, published in an equivalent form by Daniel Bernoulli in 1738. Using it correctly means remembering the four assumptions behind it:

- viscous effects are negligible
- the flow is steady
- the flow is incompressible
- it is applied along a streamline

The balancing ball

According to the Bernoulli equation, an increase in velocity can cause a decrease in pressure.



Example: flow around a bicyclist

Air flows around a bicyclist moving through still air at v_0 .

Find the pressure difference between points (1) and (2).

In coordinates fixed to the ground the flow is unsteady.

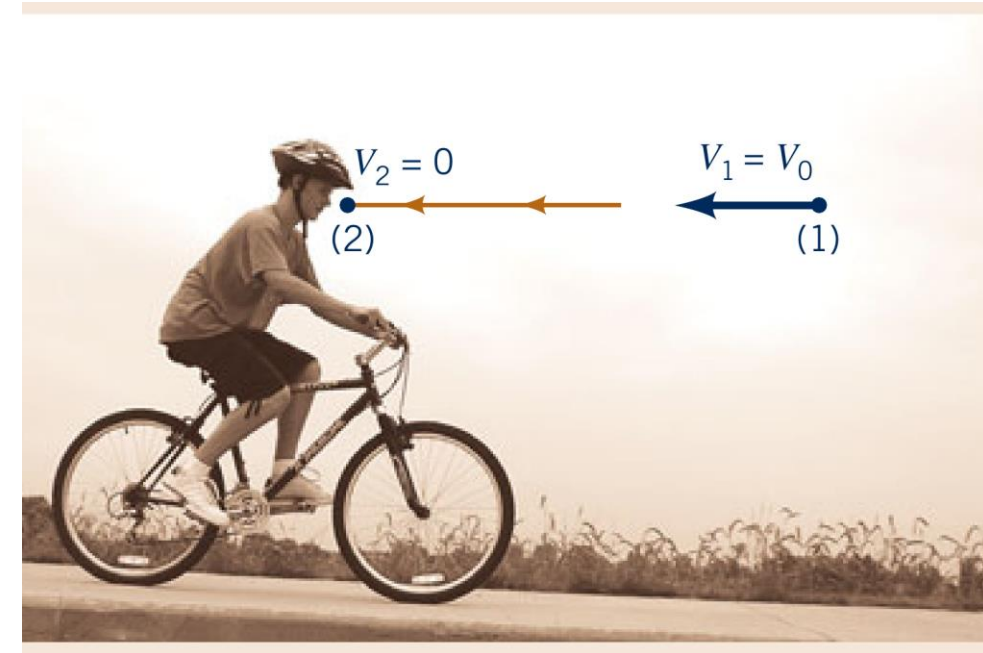
In coordinates fixed to the bike the air flows steadily towards the rider, so we use those. Then

$$p_1 + \frac{1}{2}\rho v_1^2 + \gamma z_1 = p_2 + \frac{1}{2}\rho v_2^2 + \gamma z_2$$

Point (1) is in the free stream, $v_1 = v_0$; point (2) is at the tip of the rider's nose, $v_2 = 0$; and $z_1 = z_2$. Hence:

$$p_2 - p_1 = \frac{1}{2}\rho v_1^2 = \frac{1}{2}\rho v_0^2$$

$$v_0 = 10 \text{ m/s}, \quad \rho = 1.23 \text{ kg/m}^3 : \quad p_2 - p_1 = 61.5 \text{ Pa}$$



Flow past a biker

The Bernoulli equation contains **6 unknowns**: 2 pressures, 2 velocities and 2 elevations.

To get one of them from this single equation, the other five must be known.



Video courtesy of A2 Wind Tunnel, www.A2WT.com

$F = ma$ normal to a streamline

In many flows the streamlines are nearly straight and the variation across them can be ignored.

But the low-pressure region at the centre of a tornado is explained precisely by **applying Newton's second law across its nearly circular streamlines.**

In the normal direction, with the normal acceleration v^2/R ,

$$\sum \delta F_n = \frac{\delta m v^2}{\mathcal{R}} = \frac{\rho \delta \mathcal{V} v^2}{\mathcal{R}}$$

and the component of the weight normal to the streamline is

$$\delta \mathcal{W}_n = -\delta \mathcal{W} \cos \theta = -\gamma \delta \mathcal{V} \cos \theta$$

so a vertical streamline, $\theta = 90^\circ$, has no weight contribution across the flow.

To apply $F = ma$ normal to streamlines, the normal components of force are needed.

The equation of motion normal to the streamline

The net pressure force in the normal direction follows in the same way as before:

$$\delta F_{pn} = - \frac{\partial p}{\partial n} \delta s \delta n \delta y = - \frac{\partial p}{\partial n} \delta \mathcal{V}$$

$$\sum \delta F_n = \delta \mathcal{W}_n + \delta F_{pn} = \left(-\gamma \cos \theta - \frac{\partial p}{\partial n} \right) \delta \mathcal{V}$$

and with $\cos \theta = dz/dn$ this gives

$$-\gamma \frac{dz}{dn} - \frac{\partial p}{\partial n} = \frac{\rho v^2}{\mathcal{R}}$$

A curved path is produced by the appropriate combination of pressure gradient and weight across the streamline.

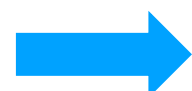
A larger speed, a larger density or a tighter radius all demand a larger imbalance.

Weight and/or pressure can produce curved streamlines.

Curved streamlines and the tornado

If gravity is neglected, as is usual for gas flows, or if the flow is horizontal, our equation reduces to:

$$\frac{\partial p}{\partial n} = -\frac{\rho v^2}{\mathcal{R}}$$

 **The pressure increases with distance away from the centre of curvature**

The positive n direction points towards the inside of the curve, so $\partial p / \partial n$ is negative.

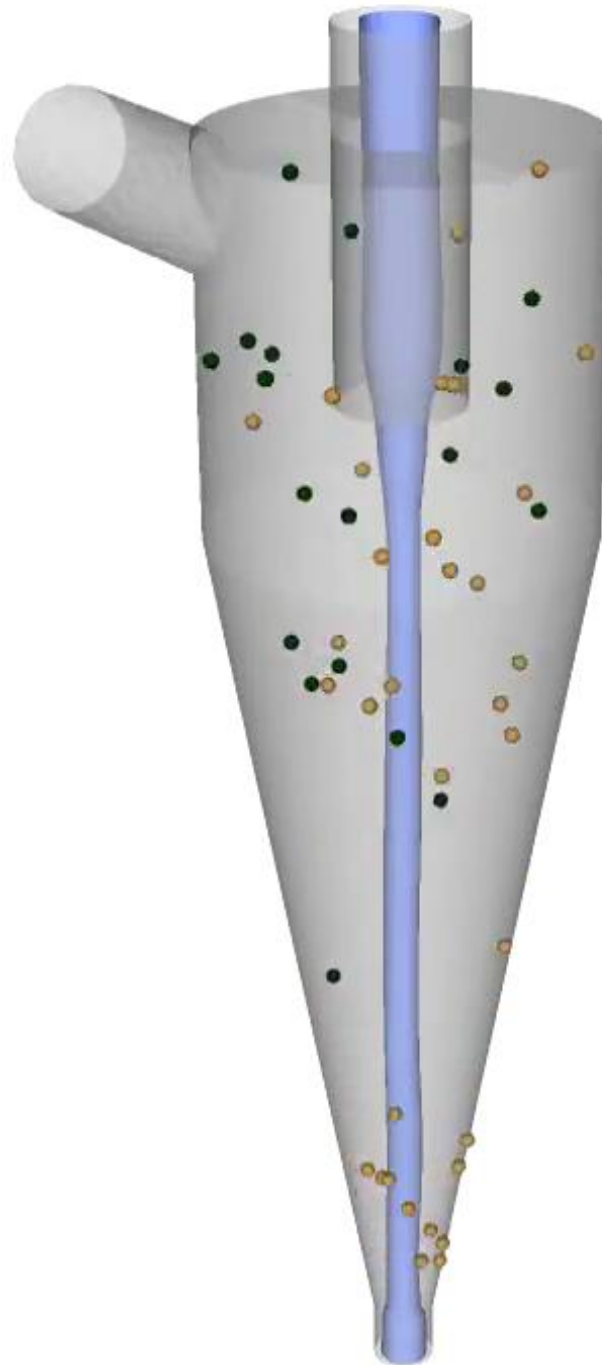
This is why the pressure outside a tornado (ordinary atmospheric pressure) is larger than the dangerously low pressure near its centre: **that difference is what balances the centrifugal acceleration of the curved motion.**

Integrating across the streamlines for constant density gives

$$p + \rho \int \frac{v^2}{\mathcal{R}} dn + \gamma z = \text{constant across the streamline}$$

The hydrocyclone separator

A hydrocyclone applies centrifugal force to a liquid mixture to separate its heavy and light components. It is a closed vessel that turns the incoming fluid motion into a rotary one.



The aircraft wing tip vortex

Many important flows have curved streamlines. The swirling motion of tornados, water spouts and water draining from a bath tub are typical examples.



Video courtesy of NASA Dryden Flight Research Center

The free vortex

For flow with curved streamlines, the centrifugal acceleration makes the pressure on the outside of the bend greater than on the inside.



Tornado video courtesy of NOAA National Severe Storm Lab; video still © B. R. Munson

A **free vortex** is a type of rotational fluid flow where the fluid rotates without any external torque or applied energy, relying instead on initial momentum or gravity.

Physical interpretation

The 2 results of the previous sections are:

$$p + \frac{1}{2}\rho v^2 + \gamma z = \text{constant along the streamline}$$

$$p + \rho \int \frac{v^2}{\mathcal{R}} dn + \gamma z = \text{constant across the streamline}$$

Three processes appear in them:

- **mass times acceleration**, in the ρv^2 term;
- **pressure**, in the p term; and
- **weight**, in the γz term

Integrating the equation of motion along the streamline is the work–energy principle of dynamics:

The work done on a particle by all the forces acting on it is equal to the change in its kinetic energy.

The Bernoulli equation is a mathematical statement of that principle. As the particle moves, both gravity and pressure do work on it.

The Bernoulli equation in terms of heads

Dividing each term of the streamwise equation by the specific weight γ gives every term the dimension of a length:

$$\frac{p}{\gamma} + \frac{v^2}{2g} + z = \text{constant on a streamline}$$

$$\underbrace{\frac{p}{\gamma}}_{\text{pressure head}} + \underbrace{\frac{v^2}{2g}}_{\text{velocity head}} + \underbrace{z}_{\text{elevation head}} = \text{constant}$$

- **the pressure head**, p/γ , is the height of a column of the fluid that would produce the pressure p
- **the velocity head**, $v^2/2g$, is the distance the fluid must fall freely from rest to reach the speed v
- **the elevation head**, z , is the potential energy per unit weight

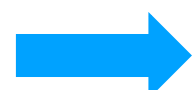
*The Bernoulli equation can be written in terms of heights called **heads**.*

Straight streamlines

When a fluid particle travels along a curved path a **net force towards the centre of curvature is needed**, and it may come from gravity, from pressure, or from both.

In many flows the streamlines are nearly straight, $R \rightarrow \infty$.

Then, the centrifugal term disappears from the normal equations and the pressure variation across the streamlines is caused by gravity alone — even though the fluid is moving.

 The pressure variation across straight streamlines is hydrostatic.

This is what lets us to drill a small hole in the wall of a pipe, connect a gauge, and read the pressure of a fluid that is rushing past at **10 m/s**.

If the streamlines at that wall were curved, the reading would be wrong.

Lecture summary

We have applied Newton's law $F = ma$ to a fluid particle in a steady, inviscid, incompressible flow: first along the streamline, and then across it (normal to it).

$$\mathbf{F} = m \mathbf{a}$$

Newton's second law

$$a_s = v \frac{\partial v}{\partial s}$$

along a streamline

$$a_n = \frac{v^2}{\mathcal{R}}$$

across streamlines

$$\begin{aligned} -\frac{\partial p}{\partial s} - \gamma \sin \theta \\ = \rho v \frac{\partial v}{\partial s} \end{aligned}$$

the equation of motion

$$\begin{aligned} p + \frac{1}{2}\rho v^2 + \gamma z \\ = \text{constant} \end{aligned}$$

the Bernoulli equation

$$\frac{\partial p}{\partial n} = -\frac{\rho v^2}{\mathcal{R}}$$

curved streamlines

$$v \mathcal{R} = \text{constant}$$

the free vortex

$$\frac{p}{\gamma} + \frac{v^2}{2g} + z = H$$

in terms of heads